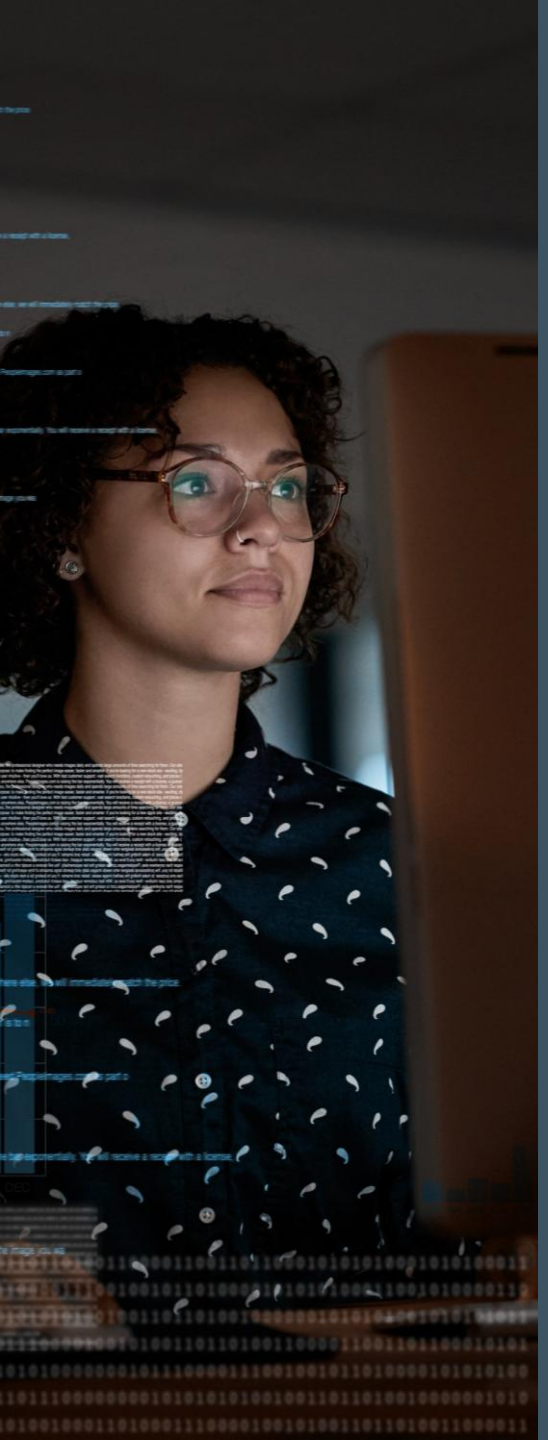


PMP® EXAM PREP BOOTCAMP

Session 5

PMI
Authorized Training Partner

ATTENDENCE TRACKING

Percipio Users:
Name is based on your log in information in Percipio

Using Zoom:
Enter your first and last name

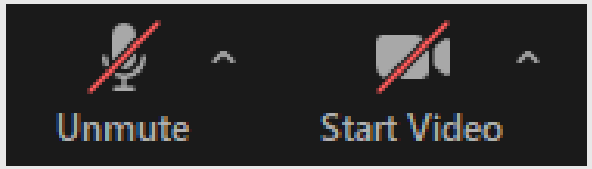
BREAKS

Yes! We will have periodic breaks.

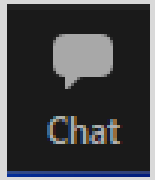
For attendance purposes, please stay logged in during all breaks.

WAYS TO PARTICIPATE

We are saving everyone's bandwidth usage by disabling cameras and microphones



Find the **Chat option** in your Zoom command bar



Change the To: field in the blue box to **Everyone.**

Explore the Reactions option in your Zoom command bar

This is a fun way to provide quick and easy feedback

CHAT vs Q&A

Please use the **Chat** for:

- **Greetings** before the session starts and during breaks
- Once the session starts , the chat may be closed or changed to *Hosts & Panelists Only* to minimize disruptions and focus on important information.
- The instructor may open the chat during the session for student **to respond to the instructor's questions** and create a group dialog.

CHAT vs Q&A

Please use the **Q&A** for:

- **Technical assistance** – Begin with: Percipio or Non-Percipio student
- Guidance on how to **access course material** – Begin with: Percipio or Non-Percipio
- Clarification and **questions on lecture points**, if not answered by instructor
- The Q&A may be open and closed throughout the session to allow us to address questions/issues in a timely manner.
- **Please be very patient, the support team responds to many inquiries per session**

IS LIVE ATTENDANCE REQUIRED?

- **YES,** if you are taking this training to register for the PMP exam
- You are **allowed to miss up to two sessions IF** you make up the sessions by **watching the video replays**.
- If you **miss three or more sessions**, you will need to make up the missed time in those additional missed sessions by **attending live in another 8-day cohort**.
- A **missed** session **means** you were disconnected for **more than a total of 15 mins** for the duration of the session.
- *Please see the Bootcamp Calendar for information about upcoming sessions at: http://calendar.skillsoft.com/

ACCESSING THE VIDEO REPLAYS

1. Go to: https://github.com/Skillsoft-Content/PMPReplay
2. Replays will be available within 2 business days after the session ends.
3. Click on the Excel file for the year you attended the Bootcamp. You won't see a *file open* option, but it is selected.
4. Click the *Download raw file* button on the far left-hand side.
5. Open the downloaded file using this password: pmpB00tcampReplay!

Those are zero's not the letter O. The password is case sensitive.

7. Locate and open the worksheet tab that corresponds with the bootcamp you attended
8. Make a note of the passcode.
9. Paste the provided link into your browser.
10. Complete the required registration steps
11. Input the passcode when prompted
    The password to open the Excel file is NOT the passcode to access the replay.

Note: Replays will be available for 1 year.
They are not available for download.

No limit to watch replays to study

Mapping this course to the Student Workbook

	Business Environment Lesson 1	Start the Project Lesson 2	Plan the Project Lesson 3	Lead the Project Team Lesson 4	Support Project Team Performance Lesson 5	Close the Project/Phase Lesson 6
Topic A	(1A) Foundation	(2A) Identify and Engage Stakeholders	(3A) Planning Projects	(4A) Craft Your Leadership Skills	(5A) Implement Ongoing Improvements	(6A) Project Phase/Closure
Topic B	(1B) Strategic Alignment	(2B) Form the Team	(3B) Scope	(4B) Create a Collaborative Project Team Environment	(5B) Support Performance	(6B) Benefits Realization
Topic C	(1C) Project Benefits and Value	(2C) Build Shared Understanding	(3C) Schedule	(4C) Empower the Team	(5C) Evaluate Project Progress	(6C) Knowledge Transfer
Topic D	(1D) Organizational Culture and Change Management	(2D) Project Approach	**(3D) Resources**	(4D) Support Team Member Performance	(5D) Manage Project Issues and Impediments	
Topic E	(1E) Project Governance		**(3E) Budget**	(4E) Communicate and Collaborate with Stakeholders	(5E) Manage Project Changes	
Topic F	(1F) Project Compliance		**(3F) Risks**	(4F) Training, Coaching and Mentoring		
Topic G			(3G) Quality	(4G) Manage Conflict		
Topic H			(3H) Integrate Plans			

LESSON 3

PLAN THE PROJECT

- Planning Projects
- Scope
- Schedule
- Resources
- Budget
- Risks
- Quality
- Integrate Plans

Version 3.2 | 2023 Release

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Learning Objectives

- Explain the importance of a project management plan.
- Provide an overview of scope planning in both predictive and adaptive projects.
- Provide an overview of schedule planning in both predictive and adaptive projects.
- **Discuss resource planning for a project, including human and physical resources and the role of procurement.**
- **Determine the budgeting structure/method for a project**
- **Explain the importance of tailoring a budget.**
- **Identify strategies for dealing with risks and risk planning.**
- **Assemble a toolkit of possible responses to risks.**
- Define quality and how it relates to the outcomes and deliveries for a project.
- Discuss the importance of integrating project management plans and tailoring a change management process.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Resources

TOPIC D

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Resources

People and Equipment

- Value and empower internal human resources, yet
- Leverage external sources to ensure you have the best team and equipment possible!

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Resource Management Plan*

- **Identify resources -** People and equipment
- **How to acquire them**
- **Peoples' roles and responsibilities**
  - Role – A person's function in a project
  - Authority - Rights to use resources, make decisions, accept deliverables.
  - Responsibility - Assigned duty
  - Competencies and skills required
- **Project Organization Chart** – (Visual with resource categories and reporting relationships)
- **Project team resource management** – Guidance on how to define, select, manage and release resources
- **Training** - Strategies and requirements
- **Team development methods**
- **Resource controls** - Methods for ensuring non-human-resources are available as needed
- **Recognition plan**

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Assign Resources and Allocate Responsibilities

- Assign team members to project
- Decide roles and responsibilities
- Create team directory, organization chart and the schedule

Project schedules, resource assignments and budgets are all interrelated and can be created at the same time.

- Tailor responsibilities according to team, needs and project approach
- Consider technical and “soft” skills:
  - Experience, knowledge, skills
  - Attitude
  - Global/regional factors

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Use Resource Calendars*

- Document resource availability (people, equipment, material, etc.) during a planned activity period.
- Use when estimating project activities and understanding dependencies
- Specifies when, and for how long, identified team and physical resources will be available during the project
- Progressively elaborate and update it throughout the project

Resource calendars *can be used in any kind of project!*

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Responsibility Assignment Tools

Responsibility assignment matrix (RAM) or **RACI chart**:

- Designates types of accountabilities assigned to resources or stakeholders
- Keeps information visible

RESPONSIBLE *A team member*	• Performs work to complete the task or create the deliverable • Every task has at least one responsible person
ACCOUNTABLE *On the team (leadership/ management)*	• Delegates and reviews the work involved in a project • Ensures the responsible person/team knows project expectations and completes work on time • Each task has only one accountable person
CONSULTED *Stakeholders*	• Provides input and feedback on project work • Not every task or milestone needs a consulted party *Consider all stakeholders, but invite only necessary input*
INFORMED *Usually not project decision makers*	• Needs to be informed of project progress because their work might be affected, but don't need details

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Adaptive Resource Planning

Quiz

Which of these are true? (Choose several)

- Teams self-organize to distribute work. ***TRUE***
- Adaptive teams never have a leader. ***FALSE***
- Team members are a mix of generalists and specialists. ***TRUE***
- Team members should be T-shaped. ***TRUE***

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Filling Resource Needs

Make or Buy? Borrow?

External sourcing considerations:

- What is the impact on cost, time or quality?
- Is there an ongoing need for the specific skill set?
- How steep is the learning curve?
- Are required resources available within the organization?
- Would outsourcing allow the team to focus?

Use a **make-or-buy analysis** to make the best decision for your team.

Make-or-buy decisions are part of a procurement strategy.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Plan the Procurement Strategy

- Prerequisite OPAs
- Acquisition method
- Contract types
- Procurement phases

- Work with organization's finance or procurement department
- Use pre-approved vendors before requesting a new vendor
- Observe purchase amount limits per signatory — i.e. contracts valued over a certain threshold must be co-signed
- Use defined bidding process and templates
- Require RFPs for contracts valued over a certain threshold
- Follow escalation procedures for approval of spending limits
- Pay contracts at a defined time – e.g., upon completion of work or at the end of a project, with net payment terms

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Procurement Management Plan*

- Specifies the types of contracts that will be used
- Describes the process for obtaining and evaluating bids
- Mandates standardized **procurement documents**
- Describes how providers will be managed

Your organization's procurement function will be involved in developing this plan. Work with them closely and use the correct procurement documents to avoid problems.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Procurement Documents

Bid and Proposal Activities

- **Statement of Work (SOW):** Details of work required
- **Request for quotation (RFQ)**: Bid/tender or quotation, including only cost
- **Invitation for Bid (IFB)**: Buyer requests expressions of interest in work
- **Request for information (RFI)**: Buyer requests more information from seller
- **Request for proposal (RFP)**: Buyer-issued statement of work required
- **Expression of Interest (EOI)**: Seller-issued expression of interest in work

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Formal Procurement Processes

RFPs, Bidder Conferences

Organizations in highly regulated industries or government

Or, if a project needs specialist work or wants to find the best quality available.

Use RFPs, **bidder conferences**, and formal processes to ensure **all prospective vendors have a clear and common understanding of the procurement**

Work closely with the procurement officer or department

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Source Selection Criteria*

Work with external resources whose values, skills and attributes are aligned with your project's.

- Overall or life-cycle cost
- Understanding of need
- Technical capability
- Management approach
- Technical approach
- Warranty
- Financial capacity
- Production capacity and interest
- Business size and type
- Past performance of sellers
- References
- Intellectual property rights
- Proprietary rights

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Qualified Vendors

- Are pre-approved by the organization
- Have a history of work with the organization
- Are often “preferred” because they are proven, and their accounts are already set up

Look in the lessons learned repository to find historical data about vendors.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Contracts*

Negotiate Productive Relationships

Contracts:

- Legalize working agreements
- Give structure to working relationships
- Further collaboration with partners
- Consider risks associated with contract types
- Deliver benefits to the buyer - different benefits by type
- Can be tailored for the partnership

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Contract Types (1 of 3)

Cost-reimbursable contracts - *For projects with expected, significant scope changes*

Involves payments (cost reimbursements) to the seller for all legitimate actual costs incurred for completed work, plus a fee (seller profit)

Cost plus fixed fee (CPFF)	• Reimburses seller for all allowable costs for performing contract work; fixed-fee payment calculated as a percentage of the initial estimated project costs. • Fee amounts do not change unless the project scope changes.
Cost plus incentive fee (CPIF)	• Reimburses seller for all allowable costs for performing contract work; predetermined incentive fee based for achieving contract-specified performance objectives. • Shares costs between buyer and seller if final costs are less or greater than the original estimated costs • Bases cost sharing on a pre-negotiated cost-sharing formula — e.g., an 80/20 split over/under goal costs
Cost plus award fee (CPAF)	• Reimburses seller for all legitimate costs • Bases majority of fee on satisfying subjective performance criteria defined and incorporated into the contract • Determines fee based on buyer's assessment of seller performance and not subject to appeals

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Contract Types (2 of 3)

Fixed-price contracts – sets a fixed total price for a defined product, service, or result; used when requirements are well defined and no significant scope changes are expected.

Firm fixed price (FFP)	Price of goods set at beginning; won't change unless scope changes
Fixed price incentive fee (FPIF)	• Gives buyer and seller flexibility • Allows for deviation from performance — i.e., financial incentives tied to achieving agreed-upon metrics (cost, schedule, awesomeness) • Sets price ceiling; any further costs charged to seller
Fixed price with economic price adjustments (FPEPA) ***Pre-approved vendors or international payments***	• Allows for special provisions for predefined final adjustments to the contract price — e.g., inflation, cost increases (or decreases) for specific commodities

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Contract Types (3 of 3)

Time and materials contracts

- Also called “time and means”
- Combine aspects of both cost-reimbursable and fixed-price contracts
- Used when a precise scope or statement of work is unavailable
- Used often for augmenting staff, acquiring experts or gaining external support

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

“Agile” Contract Types

Multi-tiered structure	• Create a master service agreement to capture fixed items - e.g., warranties, arbitration • List variable items in a schedule of services - e.g., service rates, product descriptions • Use a SOW to itemize dynamic items - e.g., scope, schedule, budget
Emphasize value delivered	• Structure milestone and payment terms based on value derived at milestones • Focus on the value of feedback in product development
Fixed-price increments	Decompose scope into smaller, fixed-price micro-deliverables (user stories), giving customer more control over how the money is spent and limiting the supplier’s financial risk.
Not-to-exceed time and materials	• Limit budget to fixed amount, allowing customer to add ideas by removing existing ones • Monitor work to avoid overage (or add contingency hours)
Graduated time and materials	• Connect quality and timely delivery of work (use DoD) to financial award – reward for early and reduce for late delivery
Early cancellation option	• Enable flexible delivery of scope, using DoD - e.g., if partial scope delivery satisfies customer, contract can be cancelled for a fee
Dynamic scope option	• Gives option to vary scope and fund innovation at specific points while limiting supplier risk • Vary scope at specific points to adjust features and innovate
Team augmentation	• Embed supplier’s services directly into the customer organization; fund team instead of scope

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Components of Contracts

- Description of work - deliverables and scope
- Delivery date and schedule information
- Identification of authority, where appropriate
- Responsibilities of both parties
- Management of technical and business aspects
- Price and payment terms
- Provisions for termination
- Applicable guarantees and warranties
- Intellectual property
- Security, confidentiality, data privacy

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

ECO Coverage

1.6 Build a team

- Deduce project resource requirements (1.6.2)

2.11 Plan and manage procurement (resources)

- Define resource requirements and needs (2.11.1)
- Communicate resource requirements (2.11.2)
- Manage suppliers/contracts (2.11.3)
- Plan and manage procurement strategy (2.11.4)
- Develop a delivery solution (2.11.5)

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Budget

TOPIC E

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Budget Planning

Overview

Consider:

- Cost as well as value
- Organization and stakeholder attitudes towards budget and costs

Create budget in accordance with project life cycles:

Begin with fixed budget and amend with change control process

Hybrid approaches add adaptability around surety

*Use **burn rate***

Agile teams collaborate with stakeholder partners and finance stakeholders to suggest incremental budgeting approaches (agile mindset)

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Predictive Budget Planning

- Create a **cost management plan**
- Employ **estimating techniques** to assign costs to activities
- Tailor a **cost baseline**
  - Is used to monitor and measure cost performance throughout the project (compares with actual results)
  - Includes budget contingencies to address identified risks
  - Can be changed only through formal change control procedures

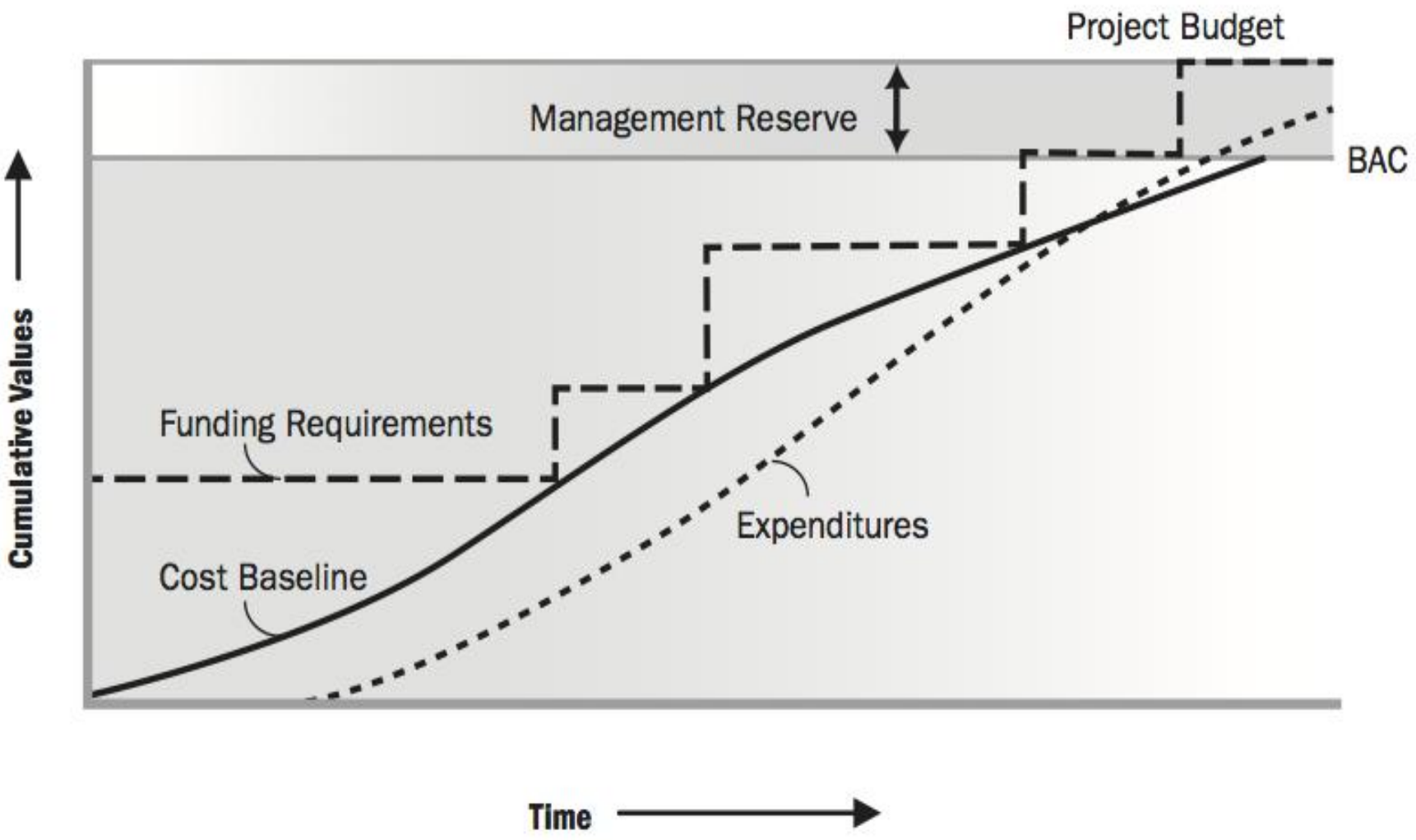
The **budget at completion (BAC)** is the highest point on the cost baseline. The BAC is the sum of all budgets established, or the value of total planned work.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Check with Organization

Funding Limit Reconciliation

- Compare planned project expenditure against funding limits
- Align work/expenditures on the schedule to level the rate of expenditures



©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

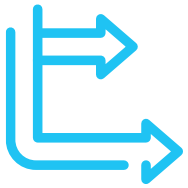
Historical Data

Start with What's Known

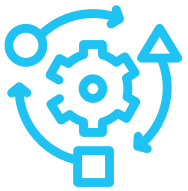
- Check lessons learned repository for budgets, estimates from previous, similar projects or data from the last iteration
- Look for valuable cost-estimating information - both successes and shortcomings
- Use analogous and estimating techniques, based on similar situations

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Resource Costs



- Match project need to resource attributes (availability, experience, knowledge/skills, attitude)
- Create initial estimate based on average rate
- Modify as needed



- Assign a blended rate
- Estimate points (effort) using planning poker or affinity diagram to find the number of user stories that can be completed based on team velocity
- Use a simple formula to estimate the cost per point:
  - Σ (loaded team salaries for period n) / points completed in interval n
- Use a formula to estimate budget:
  - (Cost per point * total point value of items to be completed) + other expenses = forecast budget

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Estimate Costs

Estimate the cost for each activity or work package in a project.

Cost estimates should include:

- Direct labor
- Materials
- Equipment
- Facilities
- Services
- Information technology
- **Contingency reserves**

Use:

- Rough order of magnitude (-25 to +75%)
- Definitive Estimate (-5 to +10%)
- Phased estimate

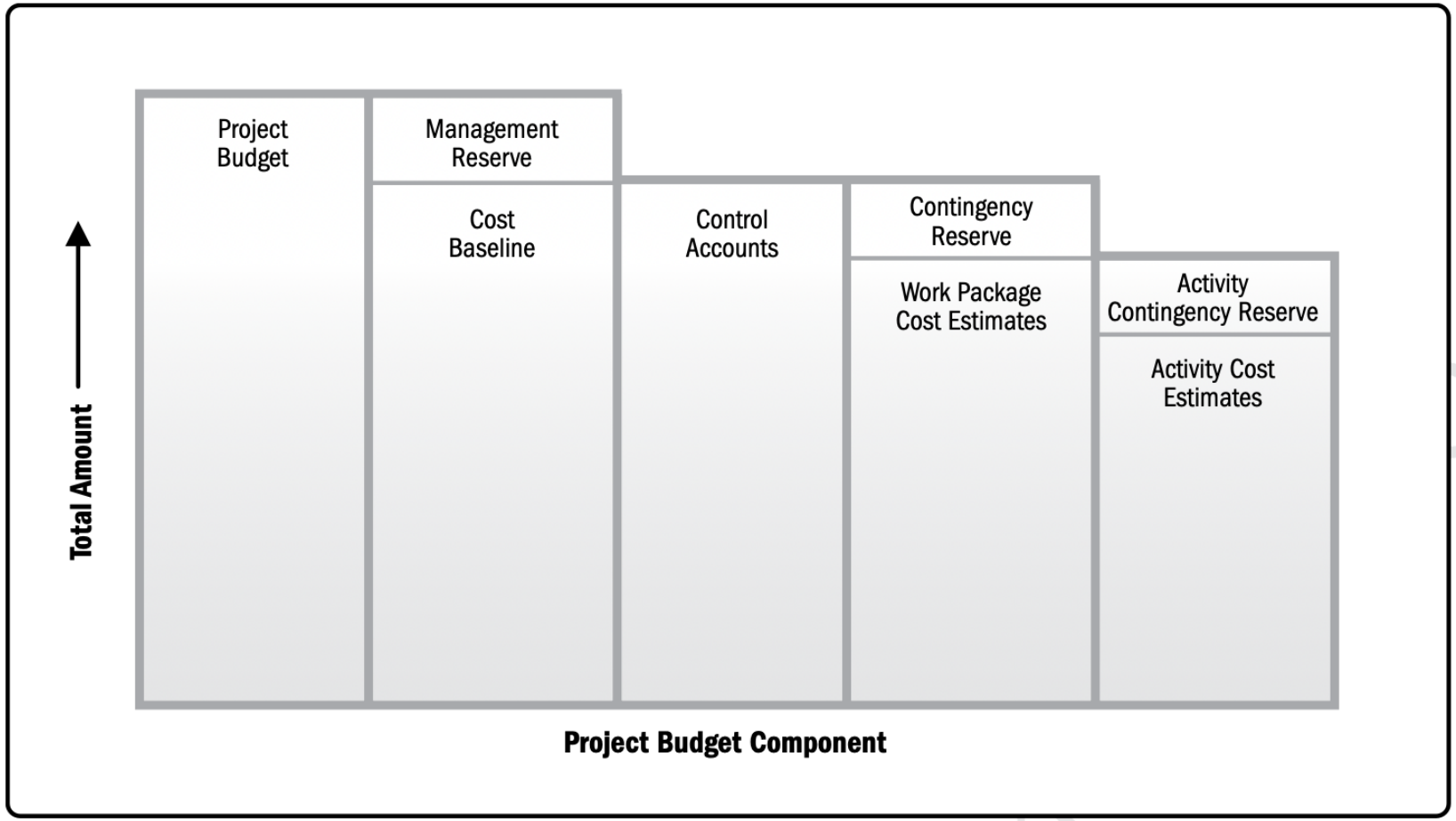
Expecting the scope to change?

Use lightweight estimation methods for high-level estimating.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Project Budget

- Use the bottom-up approach to aggregate activity costs, work package costs and cost baseline
- Include **contingencies** to support risk management



©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Adaptive and Hybrid Budget Planning

Guidelines/Example

- Focus on short-term budgeting and metrics versus long-term
- Set time periods for work and prioritize work within those time periods.
- Base cost on the resources used for that time period

Examples

- Estimate budget based on current data, plus a forecast algorithm that is based on historic data or expert guidance — e.g., lean or Kanban
- Use a “top-down” approach, using gross-level estimation techniques such as planning poker and affinity grouping on feature sets, then employing progressive elaboration and rolling-wave planning methods to drill down to the task level on a just-in-time basis (iteratively)
- Revise budget at sprint planning intervals

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Budget Considerations

- Estimate budget based on the length of time of the project
- Burn rate includes:
  - Number of team members
  - Blended or actual team member rates
  - Time of involvement
- Assumption of full-time team involvement
- If additional equipment or supplies are required, add them to the estimated cost

Product owner may control the budget, depending on team composition.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

ECO Coverage

2.5 Plan and manage budget and resources

- Estimate budgetary needs based on the scope of the project and lessons learned from past projects (2.5.1)
- Anticipate future budget challenges (2.5.2)
- Plan and manage resources (2.5.4)

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risks

TOPIC F

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk

Conditions of Uncertainty

- Risk originates from a wide range of known and unknown causes within and outside the business environment.
- Risk development is indicated by a **trigger condition**.
- Risks can be positive (**opportunities**) or negative (**threats**).
- If a risk becomes an **issue**, you must act!

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Project Risks

SLC Examples

Project Risks

- **Working with new vendors and building processes**
- **Supply chain issues for correct bricks**
- **Building code compliance**
- **Key stakeholder conflict**
- **Retail market changes – decline of in-store shopping**
- **Site survey shows risk of slippage from coastal erosion < 25 years**

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk

Business Context

- *Likelihood of a risk event vs. the potential impact*
- *Opportunity vs. threat*

Business risks represent an opportunity for gain or loss.

Project risk management systematically maximizes the probability of positive events and minimizes the probability and consequences of negative events.

As project uncertainty increases, the risk of rework increases; adaptive life cycles use smaller increments of work to enable **feedback** and **progressive elaboration** of scope.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Create Risk Strategy

First, understand risk parameters for the organization and the project!

How would you describe the organization/ project's **risk appetite**?

- Risk-seeking?
- Risk-neutral?
- Risk-averse?

The **risk threshold** is tied to individual and organizational risk appetites. Do you know:

- Which are too high to accept?
- Which are low enough to just be accepted?

- What criteria determines inclusion in the **risk register?**

Management Guidelines

- *Use qualitative (high, medium, low, etc.) or quantitative (numerical) ratings*
- *Set a maximum risk exposure level that can be managed without escalation*

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Define/Refine Risk Management Approach

Set initial risk strategy, then define and refine it!

Factor in project characteristics:

- Size
- Complexity
- Importance
- Development approach

Create a **risk management plan**!

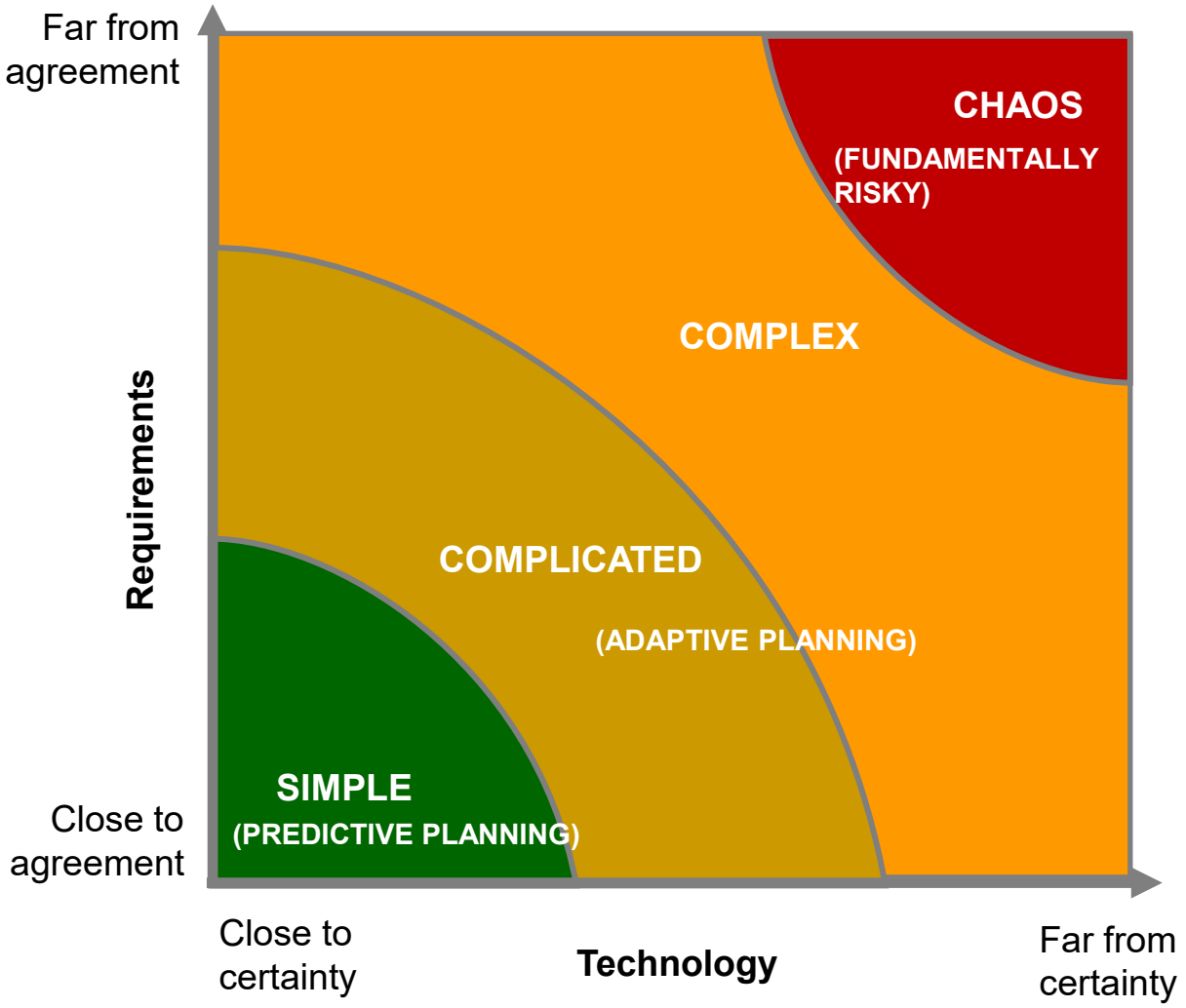
In the plan:

- Risk strategy
- Methodology
- Roles and responsibilities
- Funding
- Timing
- Risk categories
- Stakeholder risk appetite
- Definition of risk probability and impact
- Probability and impact matrix
- Reporting formats
- Tracking documents

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Inherent Risk

- Agile projects include risks in user stories and as part of backlog work items
- Teams discuss risks at planning meetings, during the normal course of work
- Teams place risks in a **risk register,** use **information radiators** to ensure visibility and a **backlog refinement** process that includes constant risk assessment



©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk Identification Techniques

*Use a **prompt list** to evaluate the external environment for risks.*

Data Gathering and Analysis

- **Risk breakdown structure (RBS)**
- Brainstorming
- Nominal group technique
- SWOT analysis
- **Affinity diagram**
- Assumption analysis
- Document review
- **Delphi technique**
- Monte Carlo simulation (larger organizations)

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk Breakdown Structure

Uses typical categories, such as:

- Technical
- Management
- Commercial
- External

RBS Level 0	RBS Level 1	RBS Level 2
0. All Sources of Project Risk	1. Technical Risk	1.1 Scope definition
		1.2 Requirements definition
		1.3 Estimates, assumptions, and constraints
		1.4 Technical processes
		1.5 Technology
		1.6 Technical interfaces
	2. Management Risk	2.1 Project management
		2.2 Program/portfolio management
		2.3 Operations management
		2.4 Organization
		2.5 Resourcing
		2.6 Communication
	3. Commercial Risk	3.1 Contractual terms and conditions
		3.2 Internal procurement
		3.3 Suppliers and vendors
		3.4 Subcontracts
		3.5 Client/customer stability
		3.6 Partnerships and join ventures
	4. External Risk	4.1 Legislation
		4.2 Exchange rates
		4.3 Site / facilities
		4.4 Environmental / weather
		4.5 Competition
		4.6 Regulatory

Example RBS

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Assess Risks

Qualitative *then* Quantitative

Perform the subjective **qualitative assessment** first.

Prioritize risks for further analysis by assessing and combining their probability of occurrence and impact in a **probability/impact matrix**.

Then, if further support is required, use a **quantitative assessment**.

Not every risk needs quantitative assessment.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Create Risk Probability and Impact Definitions

Example

+ / - IMPACT ON PROJECT OBJECTIVES

SCALE	PROBABILITY	TIME	COST	QUALITY
VERY HIGH	>70%	>6 months	>$5m	Very significant impact on overall functionality
HIGH	51-70%	3-6 months	$1m-$5m	Significant impact on overall functionality
MEDIUM	31-50%	1-3 months	$501k - $1m	Some impact in key functional areas
LOW	11-30%	1-4 weeks	$100k-$500k	Minor impact on overall functionality
VERY LOW	1-10%	1 week	<$100k	Minor impact on secondary functions
NIL	<1%	No change	No change	No change in functionality

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Probability and Impact Matrix

- Use numeric values and/or colors
- If using numbers, multiply them to give a probability impact score – this makes evaluating relative priority easier!

This is NOT a quantitative evaluation.

IMPACT (SEVERITY)

PROBABILITY (LIKELIHOOD)

	1	2	3	4	5
1	VERY LOW 1	2	3	4	5
2	2	LOW 4	6	8	10
3	3	6	MEDIUM 9	12	15
4	4	8	12	HIGH 16	20
5	5	10	15	20	VERY HIGH 25

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk Register*

Risk Description	Impact Description	Impact Level Score	Probability Level Score	Risk Score *(probability and impact multiplied)*	Trigger Condition	Planned Response	Owner
	What will happen if the risk is not mitigated or eliminated	*Rate 1 (LOW) to 5 (HIGH)*	*Rate 1 (LOW) to 5 (HIGH)*	*(IMPACT X PROBABILITY) Address highest first.*	*What indicates the risk will occur.*	*Action plan*	*Who's responsible*
Supply chain issues for correct bricks		5	1	5	Supplier notification		L. De Souza
Building code compliance		5	2	10	Pre-checks fail		K. Ayoung
Working with new vendors and building processes		3	3	9	Delays or conflict		K. Ayoung

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk List

Risk	Probability (1-10)	Impact (1-10)	Magnitude
• Working with new vendors and building processes	5	6	30
• Supply chain issues for correct bricks	5	10	50
• Building code noncompliance	5	10	50
• Key stakeholder conflict (Josie Bynoe)	4	6	24
• Retail market declining	8	10	80
• Site survey shows risk of slippage from coastal erosion < 25 years	5	3	15

Teams can add (tailor) columns for:

- Owner
- Status
- Date identified
- Date resolved
- Days active
- Resolution strategy

In addition to a risk list or a risk register, teams use information radiators and a backlog refinement process with risks added, which are discussed at various planning meetings.

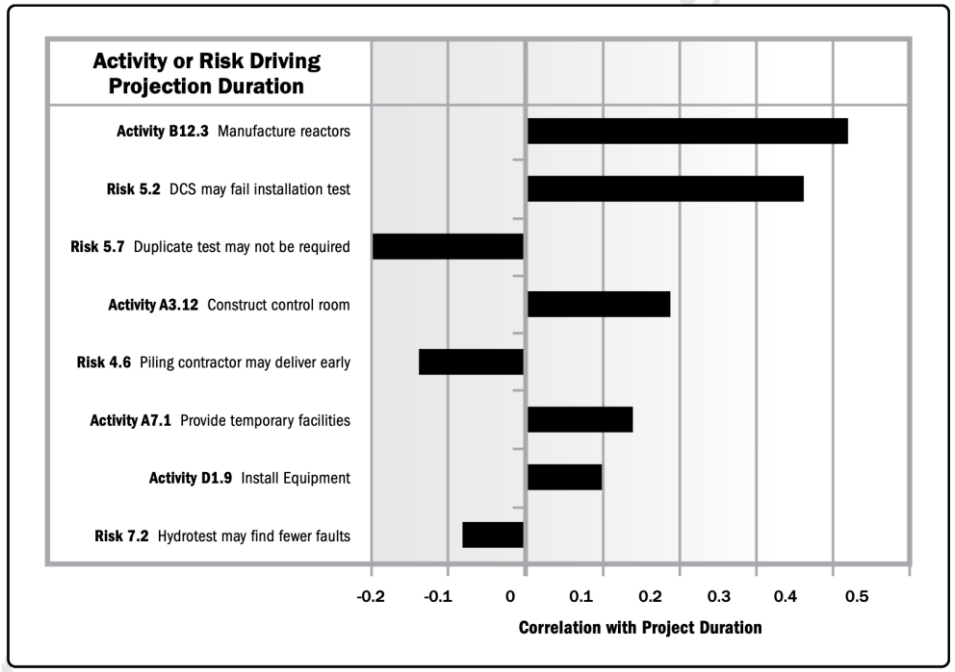
©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Quantitative Risk Analysis Methods

(1 of 2)

- **Simulations**
- **Sensitivity analysis**
- *Decision tree analysis*
- *Influence diagrams*
- *Expected monetary value (EMV)*

- **Simulations** - Use computer models to determine risk factors
  - **Monte Carlo simulations** produce a quantitative risk analysis model by using schedule and/or cost inputs to produce an integrated quantitative cost-schedule risk analysis
- **Sensitivity analysis -**
- Output is the **Tornado diagram,** a horizontal bar chart comparing relative importance of various risks, highest on top



©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Quantitative Risk Analysis Methods

(2 of 2)

- *Simulations*
- *Sensitivity analysis*
- **Decision tree analysis**
- **Influence diagrams**
- **Expected monetary value (EMV)**

Decision tree analysis

- Branches represent decisions or events, each with associated costs and risks
- The end-points of branches represent the outcome (negative or positive)

Influence diagrams

- Quality management graphical aid
- Shows elements of uncertainty caused by risks using ranges or probability distributions

Used when decision trees are too complex.

Expected Monetary Value (EMV)

- Multiply the monetary value of a possible outcome with its probability of occurrence to calculate the EMV of each branch
- Select the optimal one

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risks

Time, Cost and Life Cycle

"Predictive projects are most often affected by the impact of cost-related risks, whereas adaptive projects are affected by the impact of time-related risks."

Do you agree or disagree? Why?

Do *you think each of these typical risks is more typical of predictive or adaptive project? Can you explain why?*

Typical Risks

- Delivery date slips
- Stretched resources
- Lack of clarity
- Scope creep

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk Response

Good Practice

Risk responses should be:

- Appropriate for the significance of the risk
- Cost effective
- Realistic within the project context
- Agreed to by relevant stakeholders
- Owned by a responsible person

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Plan Risk Response

Guidelines and Terminology

- A trigger condition signals a risk can develop
- Team implements a risk response
- A **secondary risk** can arise as a direct result of the risk response implementation
- **Residual risk** can remain after risk responses have been implemented
- Have a **contingency (fallback) plan** ready in case the primary risk response fails
- The **contingency reserve (or allowance)** is the budget within the cost baseline that is allocated for identified risks and their response strategies

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

Risk Response Strategies

Prepare strategies for threats (negative) as well as opportunities (positive) and for individual project risks and overall project risk.

THREAT	OPPORTUNITY
ESCALATE	ESCALATE
AVOID	EXPLOIT
TRANSFER	SHARE
MITIGATE	ENHANCE
ACCEPT	ACCEPT

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

ECO Coverage

2.3 Assess and manage risks

- Determine risk management options (2.3.1)
- Iteratively assess and prioritize risks (2.3.2)

3.1 Plan and manage project compliance

- Determine necessary approach and action to address compliance needs (risk, legal) (3.1.6)
- Determine potential threats to compliance (3.1.3)

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

DAILY PMP BOOTCAMP SURVEY

LOOK FOR THE SURVEY LINK IN THE CHAT

Our goal is to provide the best possible Bootcamp experience for a live streaming webinar, with hundreds of participants.

For each Bootcamp session,

- Let us know **what you liked** about the experience – your comments really matter.
  - Please include a thank you **to the mentor(s)** working off camera.
- If you have **recommendations**, share those too!

We sincerely value your opinion!

Survey Scale

This Scale: 0 not at all likely- 10 extremely likely

skillsoft

On a scale of 0-10, how likely are you to recommend this bootcamp to someone else?

This Scale: 0 not at all likely - 10 extremely likely

0	1	2	3	4	5	6	7	8	9	10
○	○	○	○	○	○	○	○	○	○	○

RESOURCE MANAGEMENT PLAN

A component of the project management plan that describes how project resources are acquired, allocated, monitored, and controlled.

.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

67

RESOURCE CALENDAR

A calendar that identifies the working days and shifts for which each specific resource is available.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

68

RACI CHART

Stands for Responsible, Accountable, Consult, and Inform. A common type of responsibility assignment matrix (RAM) that uses responsible, accountable, consult, and inform statuses to define the involvement of stakeholders in project activities.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

69

MAKE-OR-BUY ANALYSIS

The process of gathering and organizing data about product/service requirements and analyzing data against available alternatives including the purchase or internal manufacture of the project.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

70

MAKE-OR-BUY DECISIONS

Decisions made regarding the external purchase versus internal manufacture of a product.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

71

PROCUREMENT MANAGEMENT PLAN

A component of the project or program management plan that describes how a project team will acquire goods and services from outside the executing organization.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

72

PROCUREMENT DOCUMENTS

Documents used in bid and proposal activities, which include the buyer's invitation for bid, expression of interest (EOI); invitation for negotiations; request for information (RFI); request for quotation (RFQ); request for proposal (RFP); and seller's responses.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

73

STATEMENT OF WORK (SOW)

A narrative description of products, services, or results to be delivered.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

74

REQUEST FOR PROPOSAL (RFP)

A type of procurement document used to request proposals from prospective sellers of products or services. In some application areas, it may have a narrower or more specific meaning.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

75

BIDDER CONFERENCES

The meetings with prospective sellers prior to the preparation of a bid or proposal to ensure all prospective vendors have a clear and common understanding of the procurement. Also called vendor conferences, pre-bid conferences, or contractor conferences.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

76

SOURCE SELECTION CRITERIA

A set of attributes, desired by the buyer, which a seller is required to meet or exceed to be selected for a contract.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

77

CONTRACT

A mutually binding agreement that obligates the seller (supplier) to provide the specified project or service or result and obligates the buyer to pay for it.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

78

BURN RATE

The rate at which the project consumes financial resources, representing negative cash flow. Burn rates are often used by agile projects to budget costs for planned iterations / sprints / increments.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

79

COST MANAGEMENT PLAN

A component of a project or program management plan that describes how costs will be planned, structured, and controlled.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

80

COST BASELINE

The approved version of the time-phased project budget, excluding any management reserves, which can be changed only through formal change control procedures and is used as a basis for comparison to actual results.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

81

BUDGET AT COMPLETION (BAC)

The sum of all budgets established to provide financial support for the work to be performed.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

82

CONTINGENCY RESERVE

Time or money allocated in the schedule or cost baseline for known risks with active response strategies.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

83

TRIGGER CONDITION

An event or situation that indicates that a risk is about to occur.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

84

OPPORTUNITY

A risk that, if developed, would create a positive effect on one or more project objectives.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

85

THREAT

A risk that would have a negative effect on one or more project objectives.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

86

ISSUE

A current condition or situation that may have an impact on the project objectives.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

87

BUSINESS RISK

The inherent risk in any business endeavor that carries the potential for either profit or loss. Types of business risks are competitive, legislative, monetary, and operational.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

88

RISK APPETITE

The degree of uncertainty an organization or individual is willing to accept in anticipation of a reward.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

89

RISK THRESHOLD

The level of risk exposure above which risks are addressed and below which risks may be accepted.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

90

RISK MANAGEMENT PLAN

A component of the project, program, or portfolio management plan that describes how risk management activities will be structured and performed.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

91

PROMPT LIST

A checklist for a specific category of risk. This tool is a simple series of broad risks, for example environmental or legal, rather than specific risks, such as flooding or regulatory changes. The idea is to push (prompt) the team to think and brainstorm the risks in groups and eventually prioritize the same.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

92

RISK BREAKDOWN STRUCTURE (RBS)

A hierarchical representation of potential sources of risk.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

93

AFFINITY DIAGRAM

A technique that allows large numbers of ideas to be classified into groups for review and analysis.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

94

DELPHI TECHNIQUE

A form of gathering expert opinions in which members of a group are asked or polled anonymously.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

95

PROBABILITY AND IMPACT MATRIX

A grid for mapping the probability of occurrence of each risk and its impact on project objectives if that risk occurs.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

96

RISK REGISTER

A repository in which outputs of risk management processes are recorded. As the central planning document for project risk analysis and control, the risk register contains a list of the most important risks to the project's completion. For each risk, it identifies the likelihood of occurrence, the impact to the project, the priority, and the applicable response plans.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

97

SIMULATION

SIMULATION

An analytical technique that models the combined effect of uncertainties to evaluate their potential impact on objectives.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

98

MONTE CARLO SIMULATION (RISK ANALYSIS)

A risk management technique, which project managers use to estimate the impacts of various risks on the project cost and project timeline. Using this method, one can easily find out what will happen to the project schedule and cost in case any risk occurs. It is used at various times during the project life cycle to get the idea on a range of probable outcomes during various scenarios.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

99

SENSITIVITY ANALYSIS

An analysis technique to determine which individual project risks or other sources of uncertainty have the most potential impact on project outcomes, by correlating variations in project outcomes with variations in elements of a quantitative risk analysis model.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

DECISION TREE ANALYSIS

A diagramming and calculation technique for evaluating the implications of a chain of multiple options in the presence of uncertainty.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

INFLUENCE DIAGRAM

Used in quality management decisions. A graphical representation of situations showing causal influences, time ordering of events, and other relationships among variables and outcomes.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

102

EXPECTED MONETARY VALUE (EMV)

A quantitative method of calculating the average outcome when the future is uncertain. The calculation of EMV is a component of decision tree analysis. Opportunities will have positive values and threats will have negative values.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

103

SECONDARY RISK

A risk that arises as a direct result of implementing a risk response.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

104

RESIDUAL RISK

The risk that remains after risk responses have been implemented.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

105

CONTINGENCY PLAN

A risk response strategy developed in advance, before risks occur; it is meant to be used if and when identified risks become reality.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

106

CONTINGENCY RESERVE

Time or money allocated in the schedule or cost baseline for known risks with active response strategies.

©2023 Project Management Institute, Inc. All rights reserved.
This material is being provided as part of a PMI® course.

107